







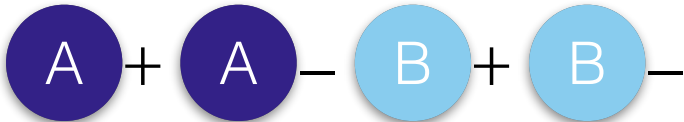
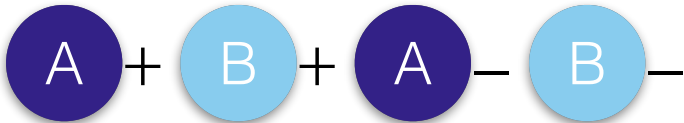
4

3



Perminituted

upd(+ -), log(+ -)



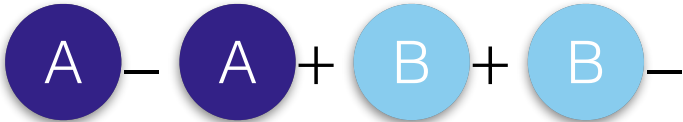
order



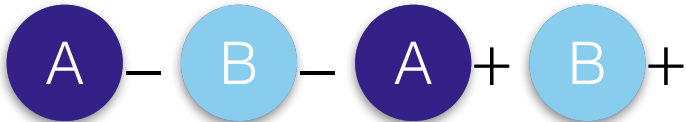
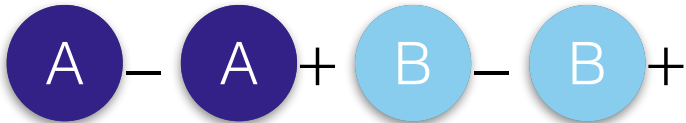
$\text{upd}(+-), \log(-+)$



upd(-+), log(+ -)



upd(-+), log(-+)













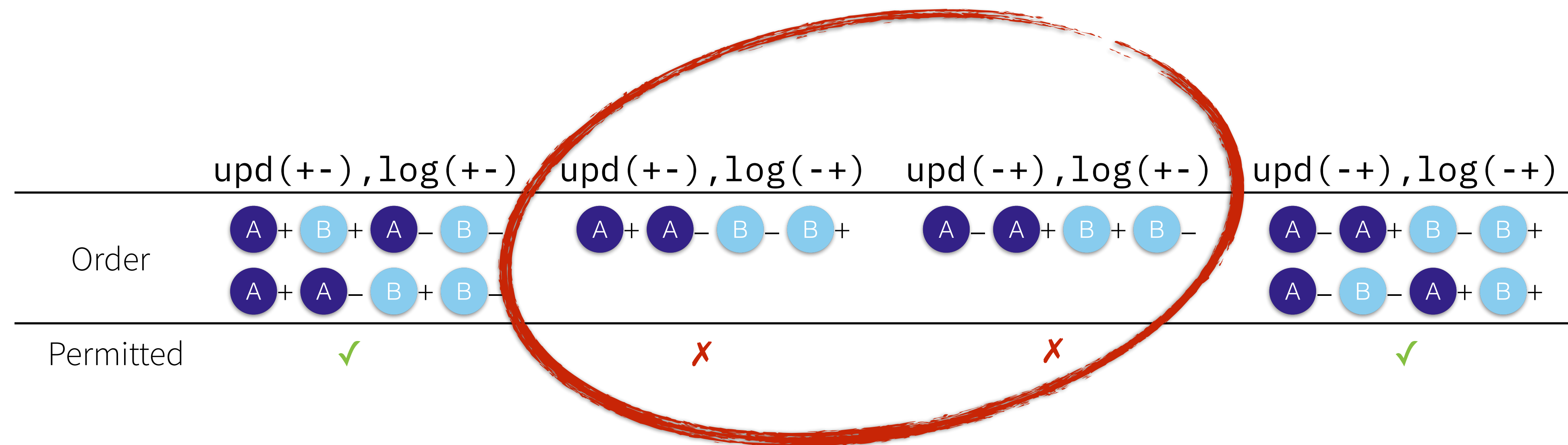


But some orders are the point!

```
def update(account: cown[Account], amount: int) {  
  when(account) {    
    account.balance += amount  
    val id = account.id()  
    when(log) {  log.record(id, amount) }  
  }  
}
```

# But some orders are the point!

```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    account.balance += amount  
    val id = account.id()  
    when(log) { B log.record(id, amount) }  
  }  
}
```





# Ordering Design Insights

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**Intrinsic** — Program structure determines which orders are allowed

**Incidental** — Scheduling and runtime must not